

for US\$ 0.25. Game developers can add, say, 100 upgraded features. We, as players, do not think too much before buying something worth US\$ 0.25. Hence, more of us end up buying an item and upgrading it without knowing how much we end up spending on it.

However, traditional payment methods do not allow these micro-transactions. Payment channels like PayPal charge US\$ 0.35 + 2.5 per cent of the transaction amount. Thereby, true micro-transaction is a myth in the gaming industry right now.

Developers are cheated. Credit card companies, especially in western countries, are extremely good in protecting their customer rights. If you have any issues regarding a purchase on your card, you are free to raise a dispute and ask for your money back. In that case, there is very little a game developer can do.

Moreover, game developers have to pay a penalty for every disputed transaction. Because of this, the gaming industry loses millions of dollars in customer support each year.

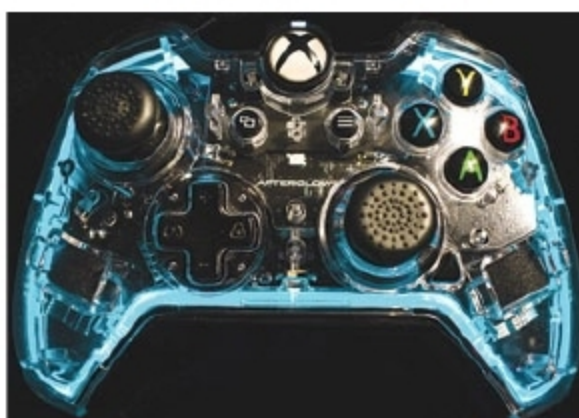
Blockchain to the rescue

Blockchain technology has the potential to end all of the above problems. Games based on blockchain technology offer solutions for game developers as well as players. There are infinite possibilities that blockchain has to offer. Some benefits are given below.

True ownership of virtual assets. In blockchain games, virtual assets belong to you, since you bought these. Creators, too, cannot take these assets away. These virtual assets may be stored in the form of tokens in your digital wallet. The tokens can be fungible or non-fungible. With blockchain games, you do not lose your digital assets, even if you get banned from, or quit, a game, or the company shuts down.

Inter-game compatibility. Since characters, items or *karma* of the games are stored on blockchain, it allows you to see your existing characters or items on a completely unrelated game. This could open a new era of gaming where characters from different game genres could play together in one game.

In case you are building your own game, instead of building new characters, you can use assets of an existing



Game controller

game or invite players from large game communities to instantly jump into your game with the characters they already have. This is one of the coolest features of blockchain technology.

Uniformity in virtual assets of different games. Design of blockchain games is fundamentally different from traditional ones. These games do not allow in-game currency to get generated infinitely. Presently, in some games, gold coins can be generated by killing a monster, while in others, these might be a scarce resource. Blockchain can end this infinite generation of coins in a game.

End the need for sub-servers. Blockchain technology in MMO role-playing games ends the need to have different servers, or sub-servers for a different region. Hence, there can be no blackout in a particular region due to failure in the server of that or another region.

Cheap and secured trading of virtual assets. Blockchain allows you to convert your virtual assets into tokens, making these free for buying, selling and trading at will. A record of all transactions is present on the blockchain itself, eliminating the threat of fraudulent activities.

Game characters, items and the like can be managed through smart contracts, as has been done in Cryptokitties—a game that created much hype in the blockchain industry.

Blockchain technology even has the potential to offer micro-transactions involving cryptocurrency in transactions.

Building blockchain games

Realising the potential of blockchain technology for next-generation online games, developers have already started using it across several verticals in

the gaming industry. There are several platforms for the development of blockchain games, in addition to payment gateways for purchasing such games. While some are already in the market, others are in beta mode or have raised funding through Initial Coin Offering. Every project has its own aim, but all of these together seem promising for building the future of gaming.

Blockchain games available today.

There are many blockchain-based games available in the market. Most are built on Ethereum, like Cryptokitties. The game uses ERC 721 tokens, which are non-fungible, indivisible and unique. It uses smart contracts for ownership and distribution of virtual cats. It is considered the first non-financial application of blockchain.

Other blockchain games are Ethermons, Cryptopets, Cryptopuppies and Etherarmy.

Ethereum is not the only blockchain to attract game developers. A blockchain game called Spell of Genesis is built on Bitcoin.

Payment systems. There are a number of blockchain-based platforms that are built for selling, buying and promoting online games. Some of these are GameCredits, Enjincoin and Refereum.

Gaming corporations, like Valve Corp., allow you to buy games through Bitcoin.

Platform for building blockchain-based games. Such platforms promise to offer all the features discussed above for blockchain games. These aim to simplify owning, sharing and trading of virtual assets. Platforms such as Bitguild and Chimaera are expected to launch this year. There are other platforms like Decentraland and Voxelus that offer a virtual space for blockchain-based 3D graphics and games.

Blockchain technology has already started expanding its roots in almost all verticals of the gaming world. From tokenisation of virtual goods to improving betting and payments, every field is undergoing several experiments.

Although, it has a lot to offer, there are issues of speed with blockchain, especially when it comes to MMO role-playing games. At present, we have a MMO role-playing game based on blockchain, but it is still in beta mode. **EFY**